

Khush A. Patel

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EDUCATION

JOHNS HOPKINS UNIVERSITY, WHITING SCHOOL OF ENGINEERING

Baltimore, MD

B.S. in Mechanical Engineering, Minor in Robotics | GPA: 4.00/4.00

Expected May 2029

- **Relevant Coursework:** Statics, Mechanics of Materials, Thermodynamics, CAD (SolidWorks, Creo), Digital Electronics, Signals and Systems, Python, Differential Equations, Linear Algebra, Multivariable Calculus

EXPERIENCE

Amphenol

Valley Green, PA

Development Engineering Intern, High Speed I/O Connectors

May 2026 – August 2026

- Designed 10+ precision high-speed connector components in SolidWorks, simplifying non-critical geometries across 5+ sheet metal and CNC-machined components to cut material usage by up to 15% under DFM principles.
- Conducted GD&T and statistical tolerance stack-up analyses on 5+ production drawings, selecting fit classes and evaluating MMC/LMC conditions using RSS and worst-case methods; partnered with machinists on the floor to confirm every drawing was manufacturable and assembled correctly before release.
- Root-caused unpredictable joint separation in spring-loaded bolted connections by developing nonlinear ANSYS static and dynamic models of preload and external loading, then built a hand-calculation tool letting engineers estimate separation loads 30x faster than running full FEA.

Johns Hopkins University AIRO Laboratory

Baltimore, MD

Research Assistant, Computational Aerodynamics & Controls

December 2025 – Present

- Built a sensor-based state estimation system for a UAV landing on a moving ship: using two sensors (one on the UAV, one fixed on the deck), trained a PyTorch model to estimate the surrounding turbulent flow state in real time, grounded in a 2M+ cell OpenFOAM simulation compressed 600x into control-ready reduced-order models at 95% accuracy.
- Validated that estimate confidence improved monotonically as the UAV accumulated sensor history over a simulated 5-second approach, benchmarking against baseline estimators and fixing pipeline bugs (including data leakage) before integrating the estimator into the lab's Model Predictive Control (MPC) framework.

Johns Hopkins Mars Rover Team

Baltimore, MD

Mechanical & Controls Engineer, Manipulator

September 2025 – Present

- Derived torque requirements analytically across all 4 wrist motors using statics, identifying worst-case loading at full extension to size BLDC actuators at 50% lower cost while meeting a 7 kg payload requirement.
- Used FEA to optimize the cycloidal gearbox housing for weight while staying under an 8 μm shaft deflection threshold required for proper gear mesh, cutting housing mass by 20% (253 g to 210 g).
- Validated the design with hand calculations before release to the machine shop: hand-calc deflection (7.2 μm) agreed with FEA (6.5 μm) within 10%, confirming the design for CNC machining in 7075-T6 aluminum at a 1.3 safety factor.

Amarkosha Technologies (Start-Up)

Baltimore, MD

Mechanical Lead, Product Development

July 2025 – December 2025

- Led mechanical development of a renewable electromechanical energy harvester end to end, iterating through 20+ design-and-test cycles and running transient CFD, FSI, and structural analyses in ANSYS to improve conversion efficiency 3x (30 W at 70% efficiency) while validating against T300 carbon fiber deflection limits.
- Selected for the Johns Hopkins SPARK Accelerator.

PROJECTS

4-DOF Robotic Arm: Design, Build, and Structural Validation

- Designed, 3D-printed, and hand-assembled a 4-DOF robotic arm end to end on a \$30 budget, integrating Arduino-based control, soldered wiring, and closed-loop encoder feedback.
- Validated structural margins with ANSYS FEA and hand calculations, achieving 0.0025 maximum strain under a 400 g payload.

TECHNICAL SKILLS

- **Precision Design & Manufacturing:** GD&T, RSS & Worst-Case Tolerance Stack-Up Analysis, DFM/DFA, DFMEA, Precision Machining
- **CAD & Simulation:** SolidWorks, PTC Creo, Windchill, ANSYS Mechanical, Fluent, CFX, OpenFOAM CFD
- **Rapid Prototyping & Machining:** CNC Milling, CNC Turning, Wire EDM, Sheet Metal Fabrication, FDM & Resin 3D Printing, Laser Cutting, Soldering
- **Robotics, Controls & State Estimation:** ROS 2, Kinematics, Motion Planning, PID, Model Predictive Control (MPC), Sensor-Based State Estimation
- **Programming:** Python (NumPy, SciPy, PyTorch), MATLAB, C++, Git, Linux (Ubuntu)